

Chapter 9: Exceptions

EEC 3150

Trevecca Nazarene University

1 Handling Exceptions

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Defn: exception

Exceptions are errors in a Python script detected by the interpreter

They come in two main classes:

- *Syntax errors* occur when you write code that violates the rules of Python's syntax; detected by the interpreter before execution
 - Missing a parenthesis or quote
 - Misspelled keyword
 - Using a variable before it's defined
- *Runtime errors* occur during the execution of a script; cannot be detected by the interpreter before execution
 - Dividing by zero
 - Out of bounds indexing of a list
 - Trying to read a file that doesn't exist

Unless handled properly, exceptions always halt your program's execution. How can we avoid this?

Handling Exceptions

Defn: exception handling

Exception handling (or error handling) is the process of detecting and dealing with errors in programs

Defn: the try except block

- We can use a try except block to perform error handling in Python
- This will give us a plan B in case our code throws an error
- It's similar to an if else block

```
try:  
    code that might throw an error  
except:  
    code to run in case of error
```

- The code in the try portion is run first
- If no error, then the block is finished
- If there is an error, then the except portion is run

Handling Exceptions

try except example

Handling indexing a list out of bounds

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